Minhaj Uddin Ahmad

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Minhaj Uddin Ahmad | Minhaj6 | Minhaj Ahmad

Tuscaloosa, Alabama - 35404

Education

- **University of Alabama** *January 2024 - Present*
  Ph.D. Civil Engineering Tuscaloosa, USA
- **University of Alabama** *January 2024 - December 2025*
  MSc. Civil Engineering Tuscaloosa, USA
  - GPA: 4.00/4.00
- **Islamic University of Technology** *May 2023*
  Bsc. Electrical Engineering Dhaka, Bangladesh
  - GPA: 3.78/4.00

Experience

- **University of Alabama [🌐]** *January 2024 - Present*
  Graduate Research Assistant Tuscaloosa, AL
  - Developed OpenCAMS, a co-simulation platform integrating traffic simulation, sensor simulation and C-V2X communications simulation by combining SUMO, CARLA and OMNeT++ simulators.
  - Implemented late-fusion of roadside LiDAR, Radar, Camera for proactive safety warning system.
  - Worked on implementation of GNSS spoofing detection and mitigation techniques, demonstrated on TraCR Annual Conference 2024 and USDOT FoT Summit 2024.
- **Cybernetics Hi Tech Solutions (Pvt) Ltd. [🌐]** *May 2023 - December 2023*
  Research Engineer (Embedded Systems) Dhaka, Bangladesh
  - Developed live multi-camera low-latency video streaming over radio transceivers from deployed robotic platforms.
  - Designed CAN-BUS data streaming module over Ethernet interface on STM32-H7 microcontroller for controlling robotic platforms.
- **ANTS Aerial Systems Ltd. [🌐]** *2022 - 2023*
  Engineering Intern Dhaka, Bangladesh
  - Worked on drone swarm flight firmware, based on Ardupilot flight controller firmware.
  - Worked on software-in-the-loop (SITL) simulation of automated mission-planning software, over an implementation of MAVLink protocol.

Presentations and Publications

C=Conference, J=Journal, P=Patent, B=Book Chapter

Journal Papers

[J.3] **Ahmad, M. U.**, Dasgupta, S., Rahman, M., Khan, S., Haque, M.W., Saba, S.R., Bodoh, D., Huynh, N., Zhao, L. and Ozguven, E.E., (2026). **Lessons Learned from the Real-World Deployment of Multi-Sensor Fusion for Proactive Work Zone Safety Application**. *Transportation Research Record*. DOI: 10.1177/03611981261441281

[J.2] Saba, S.R., Khan, S., **Ahmad, M. U.**, Cao, J., Dasgupta, S., Rahman, M., Haque, M.W., Bodoh, D., Zhao, L., Huynh, N., and Ozguven, E.E., (2026). **Infrastructure sensor-enabled vehicle trajectory generation using multimodal fusion for proactive safety applications at work zone.**. *Transportation Research Record*. (Accepted for publication)

[J.1] **Ahmad, M. U.**, Rahman, M., Sevim, A., Bodoh, D., Khan, S., Zhao, L., Huynh, N. and Ozguven, E.E., (2026). **Historical Prediction Attention Mechanism based Trajectory Forecasting for Proactive Work Zone Safety in a Digital Twin Environment**. *IEEE Internet of Things Journal*, Vol. 13, Issue 11, pp. 22846-22861. DOI: 10.1109/JIOT.2026.3669235

Book Chapters

[B.2] **Ahmad, M. U.**, Abrar, A., Dasgupta, S. and Rahman, M., (2025). **End-to-End Co-Simulation Testbed for Cybersecurity Research and Development in Intelligent Transportation Systems**. In book *Advances in Transportation Cybersecurity and Resilience*, World Scientific Publishing, 1st Edition, 2026. DOI: 10.1142/9789819831548_0006

[B.1] Rahman, M., **Ahmad, M. U.**, Chowdhury, M. and McClendon, J., (2025). **Data analytics for prediction of real-time traffic flow parameters**. In *Data Analytics for Intelligent Transportation Systems*, pp. 289-315, Elsevier. DOI: 10.1016/B978-0-443-13878-2.00011-4

Peer-reviewed Conference Proceedings and Presentations

[C.13] Abrar, A., Dasgupta, S., Mamun, A. A., **Ahmad, M. U.**, Rahman, M., Chowdhury, Alsharif, A. (2027). **Deployment Feasibility Analysis of Post-Quantum Digital Signatures in Safety-Critical C-V2X Communication for Urban Mobility Scenario**. In *Transportation Research Board Annual Meeting, Washington DC, USA, Jan 10-14, 2027.*

[C.12] **Ahmad, M. U.**, Dasgupta, S., Irfan, M. S., Rahman, M., Chowdhury, M., Bandi, T. N. (2027). **Treating Statewide CORS Networks as Spatially Distributed Sensors for GNSS Integrity Monitoring under Unintentional and Deliberate Threats**. In *Transportation Research Board Annual Meeting, Washington DC, USA, Jan 10-14, 2027.*

[C.11] Kong, J., **Ahmad, M. U.**, Hainen, A., Rahman, M., Jones, S., (2027). **Evaluating Roadside Responder Safety Using Camera and LiDAR: Driver Behavior Analysis and Real-Time Near-Miss Detection**. In *Transportation Research Board Annual Meeting, Washington DC, USA, Jan 10-14, 2027.*

[C.10] **Ahmad, M. U.**, Irfan, M. S., Dasgupta, S., Rahman, M., (2026). **gr-pocketsdr: Bringing PocketSDR Front-Ends to GNU Radio and GNSS-SDR Ecosystem**. In *GNU Radio Conference 2026, 'Hardware for GNU Radio' track, Raleigh, North Carolina, USA, Sep 21 – 25, 2026.* Accepted for technical proceedings and talk.

[C.9] **Ahmad, M. U.**, Zaman, Z., Sun, S., Rahman, M., (2026). **Small Language Model enabled Autonomous agent for Language-Conditioned Cognitive Radar**. In *2026 IEEE International Workshop on Machine Learning for Signal Processing, Atlanta, Georgia, USA, Sept 28th - Oct 1st, 2026.*

[C.8] **Ahmad, M. U.**, Irfan, MS., Dasgupta, S., Rahman, M., (2026). **Beyond Legacy L1 C/A: Characterizing Advanced Multi-Frequency Spoofing Against Multi-Frequency GPS Receivers for Autonomous Navigation**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1 Technical Session, Orlando, Florida, USA, September 14 - 18, 2026.* DOI: 10.33012/2026.20673

[C.7] Dasgupta, S., **Ahmad, M. U.**, Irfan, MS., Rahman, M., Larson, JD., (2026). **Resilient Urban Vehicle Navigation in GNSS-Denied Environments via Tightly Integrated Inertial Propagation, Deep Motion Inference, GIS Route Constraints, and Opportunistic Visual Landmark Corrections**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), C1: Navigation Resilience to Interference and Cyber-Attacks 1 Technical Session, Orlando, Florida, USA, September 14 - 18, 2026.*

[C.6] Irfan, MS., **Ahmad, M. U.**, Zaman, Z., Dasgupta, S., Rahman, M., (2026). **A Synchronous GNSS Spoofing Attack Model by Exploiting Basic Safety Message Transmissions for Connected and Automated Vehicles**. In *The 39th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2026), F3: Resilient and Secure PNT Against Interference and Emerging Threats 2 Technical Session, Orlando, Florida, USA, September 14 - 18, 2026.*

[C.5] **Ahmad, M. U.**, Dasgupta, S., Rahman, M., Khan, S., Haque, M.W., Saba, S.R., Bodoh, D., Huynh, N., Zhao, L. and Ozguven, E.E., (2026). **Lessons Learned from the Real-World Deployment of Multi-Sensor Fusion for Proactive Work Zone Safety Application**. In *Transportation Research Board Annual Meeting 2026*. DOI: 10.48550/arXiv.2508.01599

[C.4] **Ahmad, M. U.**, Rahman, M., Sevim, A., Bodoh, D., Khan, S., Zhao, L., Huynh, N. and Ozguven, E.E., (2026). **Historical Prediction Attention Mechanism based Trajectory Forecasting for Proactive Work Zone Safety in a Digital Twin Environment**. In *Transportation Research Board Annual Meeting 2026*. DOI: 10.48550/arXiv.2508.06544

[C.3] Saba, S.R., Khan, S., **Ahmad, M. U.**, Cao, J., Rahman, M., Zhao, L., Huynh, N. and Ozguven, E.E., (2026). **Infrastructure Sensor-enabled Vehicle Data Generation using Multi-Sensor Fusion for Proactive Safety Applications at Work Zone**. In *Transportation Research Board Annual Meeting 2026*. DOI: 10.48550/arXiv.2509.25452

[C.2] Ibrahim, M.S., Rahman, S., Hasan, M.S., **Ahmad, M. U.** and Abrar, A., (2022). **Flow-Based Path Planning for Multiple Homogenous UAVs for Outdoor Formation-Flying**. In *2022 7th International Conference on Mechanical Engineering and Robotics Research (ICMERR)*, pp. 18-26, IEEE. DOI: 10.1109/ICMERR56497.2022.10097797

[C.1] **Ahmad, M. U.**, Akib, A.A.R., Raihan, M.M.S. and Shams, A.B., (2022). **ABO3 Perovskites' Formability Prediction and Crystal Structure Classification using Machine Learning**. In *International Conference on Innovations in Science, Engineering and Technology (ICISET)*, pp. 480-485, IEEE. DOI: 10.1109/ICISET54810.2022.9775906

Open-Source Contributions

- **GNSS-SDR [🌐]** *2026-*
  Software GNSS Receiver Contributor
  - Added a new PocketSDR signal source to GNSS-SDR source code.
  - Contributed to multiple data analysis tools of GNSS-SDR.

- **gr-pocketsdr [GitHub]** *2026-*
  GNU Radio OOT Module Repository Owner
  - Implemented a GNU Radio Out-of-Tree (OOT) block which enables interfacing PocketSDR front-ends to GNU Radio ecosystem.

- **OpenCAMS [GitHub]** *2025*
  SUMO-CARLA-OMNeT++ Co-Simulator Repository Owner
  - Developed a co-simulation environment that combines three major aspects of Intelligent Transportation System (ITS) - traffic simulation (SUMO), 3D sensor simulation (CARLA), and communications simulator (OMNeT++).
  - All source code is publicly available on GitHub.

- **MRTKLIB [GitHub]** *2026*
  A modernized implementation of RTKLIB with WebUI Contributor
  - Bug fixes.

Technology Demonstration

- **USDOT Future of Transportation Summit** *2024*
  US DOT Headquarters Washington DC
  - Demonstration of secure GNSS-based Autonomous Vehicle Navigation [🌐].

- **TraCR Annual Conference** *2024*
  National Center for Transportation Cybersecurity and Resiliency Greenville, SC
  - Demonstration of secure GNSS-based Autonomous Vehicle Navigation [🌐].

Teaching and Mentorship

- **Research Experiences for Undergraduates (REU)** *2026*
  Mentored undergraduate students under NSF-REU program University of Alabama
  - Craig Kudson
  - Rais Tefiku

Skills

- **Programming Languages:** Python, C, C++, Verilog
- **Database Systems:** SQLite, MongoDB, Redis
- **Data Science & Machine Learning:** Classical Machine Learning and Deep Learning, Computer Vision, Spatiotemporal Modeling
- **Development Technologies:** Linux, Git, GitHub/GitLab, Docker/Podman, Kubernetes, Proxmox
- **Specialized Simulation:** SUMO, CARLA, OMNeT++

Leadership Experience

- **University of Alabama Student Chapter of the Institute of Transportation Engineers** *2025 - 2026*
  Vice President [🌐]
  - Organizing invited talks and programs in collaboration with Alabama Transportation Institute and student organizations.

- **IEEE IUT Student Chapter** *2021 - 2023*
  Technical Officer [🌐]

Additional Information

Languages: English (Fluent), Bengali (Native)
Interests: Embedded and Linux systems, home server and self-hosting services, amateur radio.

REFERENCES

1. **Dr. Mizanur Rahman**
   Assistant Professor,
   University of Alabama
   Email: mizan.rahman@ua.edu
   Phone: +1 205 348 1717
   Relationship: PhD supervisor